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Major Article

The influence of spatial configuration on the frequency of use of hand sanitizing stations in health care environments

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Key Words:
Hand hygiene
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Health care-associated infection
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Background: The lack of user-friendly, accessible, and visible hand sanitizing stations (HSSs) in health care environments are significant factors affecting low hand hygiene compliance rates among caregivers.
Objective: To determine whether the simulated parameters of visibility and global traffic flow score for an HSS can influence the frequency of use of that HSS.
Methods: Space syntax was used to measure virtual simulation of spatial layouts of 3 units to provide quantitative visibility and global traffic flow scores for each HSS. The frequency of use of HSSs was measured for 2 weeks in 3 units in a community hospital through electronic tracking with self-developed motion sensors. Behavioral observations were also conducted during the same period to validate hand hygiene data obtained through electronic tracking. Linear models were used to tests how much variance in use is accounted for when visibility and/or global traffic flow are included in the model.
Results: When the visibility score for an HSS increases (decrease), frequency of use of the HSS will increase (decrease) (F [5, 65] = 13.877; $P < .001$). When the global traffic flow score for an HSS increases (decrease), frequency of use of the HSS will increase (decrease) (F [5, 65] = 13.877; $P < .001$).
Conclusions: This study proposed and validated a novel approach of using space syntax simulations to predict and optimize hand hygiene behavior.

Health care-associated infections (HAIs) affect more than 1.7 million individuals annually,[1] and cost the US economy approximately $8.2 billion.[2] Although various factors contribute to HAIs, contact transmission via the hands of clinical staff is the key pathway.[3] Hand hygiene compliance (HHC) with alcohol-based handrub or soap and water is considered the top infection-prevention measure to reduce pathogen transmission in health care environments.[4] However, achieving high levels of HHC has been a challenge.[5] Some studies indicate that even small improvements in HHC can reduce HAI rates.[6]

The absence of user-friendly or intuitive facilities and resources,[7] or low accessibility and visibility of hand hygiene (HH) supplies are key barriers believed to account for low HHC, making it difficult for caregivers to follow HH procedures.[8] Several studies indicate that the frequency of HHC was increased by improving the accessibility and visibility of hand sanitizing stations (HSSs).[7] Further, infection prevention guidelines, including those by the Centers for Disease Control and Prevention and World Health Organization, recommend the allocation of HSSs in locations with high visibility and accessibility.[6,9] However, when it comes to spatial allocation of HSSs, there is currently no systematic method to guide the decision-making process.

Currently, deciding where to place HSSs in hospitals is mainly intuitive and experiential. In 2013, a study by Western Michigan University proposed a theoretical framework for a 2-stage method for the optimal placement of HSSs: determine optimal locations customized to clinical units' specific layouts and workflows and select optimal HSS locations.[10] However, to date, no automated systematic process exists to identify the workflow path to highlight optimal locations or evaluate the effectiveness of HSS placement to accommodate and adapt to constant shifts in staff workflow processes, supplies, and facility functions. The existing research has mainly utilized methods such as behavioral observations and workflow diagrams to decode the movement flow and identify the most accessible and visible locations and is lacking in quantitative or analytical approaches. The main objective of this study was to propose a foundation for developing a quantitative and automated approach to optimize the allocation and placement of HSSs

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for application in health care facilities, with the goal of maximizing adherence to HHC by environmental system design. Also, this study examined the relationship between proposed simulated values of visibility and global traffic flow based on spatial configuration and the placement of HSSs in 3 inpatient units of a community hospital and the actual frequency of the use of HSSs. By testing each of the following hypotheses, we hoped to determine whether or not the parameters of the proposed computerized assessment characterizing the spatial structure of a hospital nursing unit (visibility and global traffic flow) will predict the use of HSSs.

Hypothesis 1: When both simulated parameters of visibility and global traffic flow score for a hand sanitizing station increase (decrease), the frequency of use of HSSs will increase (decrease).

Hypothesis 2: When the simulated parameter of visibility score for an HSS increases (decreases), the frequency of use of HSSs will increase (decrease).

Hypothesis 3: When the simulated parameter of global traffic flow score for an HSS increases (decreases), the frequency of use of HSSs will increase (decrease).

Findings from this study can help to determine whether simulated spatial parameters of a unit floor plan can predict human behavior and if so, can they be used as a strategic tool to improve HHC and reduce HAI?

METHOD

Study design

This prospective cross-sectional study conducted over 2 weeks was used to test the hypothesis that spatial design as measured by computer-simulated values can predict HHC among occupants in health care environments. Visibility (independent variable 1) and global traffic flow (independent variable 2) generated from virtual simulation of spatial layouts of 3 units in a community hospital were attribute variables measured with space syntax. An attribute variable is a variable that can be measured as a given characteristic of a floor layout, but not manipulated. The frequency of use of HSSs (dependent variable) for any occupants, including caregivers, families and visitors, and patients, were measured via electronic tracking. The 3 units (cardiac intensive care, medical-palliative care, and neonatology) were chosen by convenience sampling.

Participants

The study was conducted from April 11-24, 2016, in a community hospital located in upstate New York.

Data collection

Measurement of simulated spatial layouts using space syntax

The space syntax method was used to measure simulated characteristics of the spatial layouts—quantitative visibility and global traffic flow values—for each HSS of the 3 acute care units.[11] These values indicate the simulated possibility of seeing the targeted HHS or passing it by. The spatial movement network was developed by overlaying the current location of HSS with respect to the other clinical functions on the architectural floorplans. Visibility (independent variable 1) was measured by calculating the number of spaces immediately connecting a space of origin, and global traffic flow (independent variable 2) was measured by calculating the visual distance from all spaces to all others.[11] Figures 1 and 2 provide examples of visibility and global traffic flow calculations.

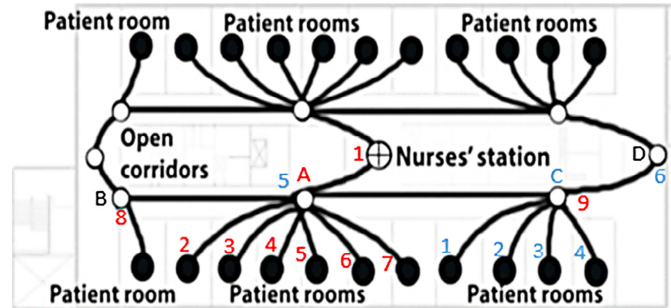

Fig 1. Visibility score of corridors. Corridor A score = 9 and corridor C score = 4.

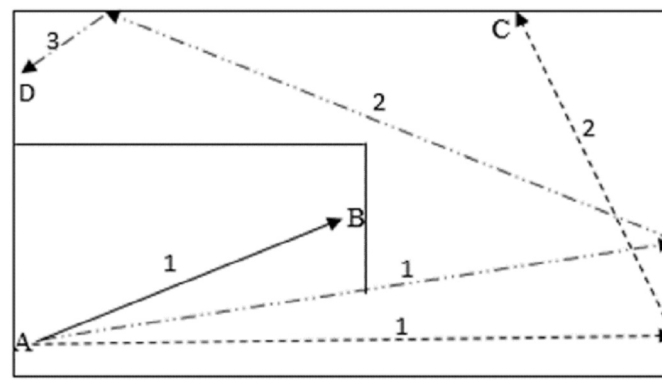

Fig 2. Determination of visual integration. Calculation of average traffic flow score from point A to points B (1), C (2), and D (3) = (1 + 2 + 3) / 3 = 2.

Electronic tracking of the frequency of use of HSSs

The frequency of use of HSSs (dependent variable) was measured from April 11-24, 2016, through electronic tracking with self-developed motion sensors. The motion sensors recorded data whenever a movement was detected as a user pushes the mechanical HSS to dispense soap or alcohol-based handrub. Due to the design of the motion sensors, only HH data from mechanical HSSs could be tracked. Seventy-one HSSs were chosen by convenience sample with 24 from cardiac intensive care, 23 from medical-palliative care, and 24 from neonatology. Sample rates were defined by the number of HSSs measured with electronic tracking / total number of available HSSs. The sampling rates were cardiac intensive care unit (23 / 44 [54.5%]), medical-palliative care unit (23 / 63 [36.5%]), and neonatology unit (24 / 68 [35.3%]). The criteria to select an HSS for convenience sampling were the HSS was located in a public space (eg, nurses workstation or outside a patient's room) and the HSS was of a mechanical design (instead of automated). The motion sensors were installed under the close supervision of permitted hospital staff from the Environmental Services Department. To control for variations in HH behaviors at different times of the day, electronic tracking that provided continuous and uninterrupted HH data over 2 weeks was used for this study. No anomaly trend (eg, outbreak of HAIs or flu season) was observed during the study duration.

Validation of electronic tracking via behavioral observations of the frequency of use of HHSs

Seventy-seven HSSs from the 7 units were sampled via behavioral observations; that is, observation of the frequency of use of HSSs among occupants. For each unit, the number of sampled HSSs were: cardiac intensive care unit (n = 25), medical-palliative care unit (n = 24), and neonatology unit (n = 28). Twenty hours of behavioral observations were conducted for each unit. To be sampled

Table 1
Descriptive statistics for visibility, traffic flow, and use

	n	Minimum	Maximum	Mean	Standard deviation
Visibility score	71	11	979	341.45	260.24
Global traffic flow score	71	2.70	12.58	6.53	2.61
Use of each hand sanitizing stations over 2 wk	71	29	1305	264.30	259.78

via behavioral observation, each HSS needs to be located in a public space (eg, nurses workstation or outside a patient's room). Observations were conducted over the morning and afternoon shifts (between 7 a.m. and 7 p.m.) and all days of the week. The start and end time of each observation session was noted to ensure that each unit was compared with the same time periods. The observation schedule was created in a way to allocate an equal period of observation times for each of the 3 units.

Data analysis

Descriptive statistics (number, minimum, maximum, and mean ± standard deviation) were run to check values of visibility, global traffic flow, and use for any anomalous values. Several past studies related to predictive models used in clinical studies considered a P value < .05 to be significant.[12,13] In this study, a P value < .05 would be considered significant, following past studies with similar assumptions.[14] A univariate general linear model was used to test our hypothesis, which suggests that when visibility and/or global traffic flow increase (decrease), the frequency of use of HSSs will increase (decrease).

The unit of analysis here was an HSS. The sum of the frequency of use of HSSs over 2 weeks was entered into the data file. To determine the reliability of data obtained from electronic tracking, Pearson correlations were run to assess the association between HH data obtained from electronic tracking and behavioral observations and there was a high level of correlation ($r = 0.981$). Hence, data from electronic tracking were reliable in predicting the frequency of use of HSSs and were used in the model. Data from electronic tracking were used in the model because this type of data collection provided us with continuous HH data over 2 weeks.

In this analysis, typical linear model assumptions were tested and we subsequently used log transformation of dependent variables (ie, frequency of the use of HSSs) in the data analysis. Scatter plots (see online resources eg, https://en.wikipedia.org/wiki/Scatter_plot) were run to assess whether relationships between variables were linear and whether particular HSSs represented outliers with respect to patterns of correlations.

Linear models were used to test the hypotheses. The first regression model tests how much variance in use is accounted for when both visibility and global traffic flow (independent variable 1 and independent variable 2) are included in the model. The second regression model tests how much variance in use is accounted for when only visibility (independent variable 1) is included in the model. The third regression model tests how much variance in use is accounted for when only global traffic flow (independent variable 2) is included in the model.

RESULTS

Space syntax was used to measure virtual simulation of spatial layouts of the 3 units to provide quantitative values of visibility and global traffic flow. Overall, visibility values ranged from 11-979 (mean, 341.45 ± 260.24) and global traffic flow values ranged from 2.70-12.58 (mean, 6.53 ± 2.61) across the 3 units (see Table 1).

Electronic data (validated by behavioral observations) on the frequency of use of HSS were collected from 71 HSSs in the 3 units: cardiac intensive care unit (n = 24), medical-palliative care unit (n = 23), and neonatology unit (n = 24). Overall, the frequency of use of HSSs ranged from 29-1,305 (mean, 264.30 ± 259.78) across the 3 units (see Table 1).

The influence of visibility and global traffic flow on the frequency of use of HHSs (model 1)

When visibility and global traffic flow were in the same 2-way regression model, the variance inflation factor was 10.092. Because the variance inflation factor > 10, it means that the visibility and global traffic flow values are highly correlated and thus cannot be in the same model (see Table 2). Hence, we were unable to support or refute hypothesis 1, which states that when the visibility and global traffic flow score for an HSS increases (decreases), the frequency of use of the HSSs will increase (decrease). Because visibility and global traffic flow scores are highly correlated with our data, it is not clear whether the higher frequency of the use of HSSs is due to visibility or global traffic flow.

The influence of visibility on the frequency of use of HSSs (model 2)

Space syntax was used to measure virtual simulation of spatial layouts of the 3 units to provide quantitative values of visibility. Overall, visibility values ranged from 11-979 (mean, 341.45 ± 260.24) across the 3 units (see Table 1).

The 2-way regression model between the visibility scores and the frequency of use of HSSs were significant (F [5, 65] = 13.877; $P < .001$) and adjusted $r^2 = 0.479$. The influence of visibility measured by simulation was significant in affecting the frequency of use of HSSs (F [1, 65] = 41.377; $P < 0.001$) (see Table A1). The results support hypothesis 2, which states that when the simulated visibility score for an HSS increases (decreases), the frequency of use of the HSS will increase (decrease).

The standardized regression coefficient was significantly different from zero for visibility ($\beta = 0.004$; $P < .001$) (see Table A2). Because the beta value on visibility score was 0.004, for every 1-unit change in visibility score, the frequency of use of HSSs changes by 0.004.

Table 2
Coefficients*

		Unstandardized coefficients		Standardized coefficients			Collinearity statistics	
Model		β	Standard error	β	t	P value	Tolerance	Variance inflation factor
1	Constant	4.746	0.405		11.713	<.001		
	Visibility	0.004	0.001	1.012	3.464	.001	0.099	10.092
	Global traffic flow	−0.154	0.114	−0.393	−1.343	.184	0.099	10.092

*Dependent variable = log use.

The influence of global traffic flow on the frequency of use of HHSs (model 3)

Space syntax was used to measure virtual simulation of spatial layouts of the 3 units to provide quantitative values of global traffic flow. Overall, global traffic flow values ranged from 2.70-12.58 (mean, 6.53 ± 2.61) across the 3 units (see Table 1).

The 2-way model between the global traffic flow scores and the frequency of use of HSSs was significant (F [3, 67] = 16.054; P < .001) and adjusted r^2 = 0.392. The influence of global traffic flow measured by simulation was significant in affecting the frequency of use of HSSs (F [1, 67] = 25.390; P < .001) (see Table A3). The results support hypothesis 3, which states that when the simulated global traffic flow score for an HSS increases (decreases), the frequency of use of the HSS will increase (decrease).

The standardized regression coefficient was significantly different from zero for global traffic flow (β = 0.379; P < .001) (see Table A4). Because the beta value on global traffic flow score was 0.379, for every 1-unit change in global traffic flow score, the frequency of use of HSS changes by 0.379.

DISCUSSION

This study aimed to examine the influence of the simulated spatial parameters of a hospital unit as measured by visibility and/or global traffic flow score of an HSS on the frequency of use of HSSs. Based on the results, the main trend revealed was that when the simulated visibility or global traffic flow score for an HSS increases (decreases), the frequency of use of HSSs will increase (decrease).

The results support the initial hypothesis (hypothesis 2) that when the visibility score for an HSS increases (decreases), the frequency of use of HSSs will increase (decrease). Visibility "measures the number of spaces immediately connecting and adjacent to a space of origin."[11] For example, the visibility score of corridor A is 9 (with immediate adjacency to 1 nurses station, 6 patient rooms, and 2 corridors [B and C]). The visibility score of corridor C is 4 (with immediate adjacency to 4 patient rooms and 2 corridors [A and D]) (see Fig 1).

The results revealed that when the global traffic flow score for an HSS increases (decreases), the frequency of use of HSSs will increase (decrease). The results support the initial hypothesis (hypothesis 3). Global traffic flow of a point is based on the number of visual steps it takes to get from that point to any other point within the global system[15] and is a good predictor of global human traffic flow.[16,17] Space syntax theory uses a mathematic equation to determine visual integration, which shows how each point is connected to all others in a space in terms of the maximum possible direction changes. For example, if the traffic flow score for point A, B, C, and D is 1, 2, and 3, respectively, the average traffic flow score for point A is (1 + 2 + 3) / 3 = 2 (see Fig 2).

The space syntax simulation method measures characteristics of a spatial floor that is by evidence linked to human behavior within that layout. For example, characteristics of the floor layout simulated by the space syntax method has predicted location, frequency, and nature of informal interactions among doctors and patients in a hospital,[18] predicted possibilities of rise in land values,[19] occurrences of crimes,[20] or in traffic and distribution of pedestrians in streets that are important for retail businesses.[21,22] This study introduces space syntax methods and analysis of the unit floor layout as a method of predicting and optimizing behavior desired for preventing infection transfer when it comes to compliance with HH.

This study has the following limitations: The proposed method only measured the influence of spatial configuration and did not provide any insight on other environmental, social, and psychological variables or protocols, policies, and education with regard to HHC. Electronic HH data were collected over 2 weeks in 3 inpatient units with varying levels of acuity within a community hospital. Thus, time of the year, types of inpatient units, and health care facilities should be studied to increase the generalizability of the findings. Furthermore, this study was a cross-sectional study conducted over 2 weeks. Due to cost constraints, the research team was unable to develop motion sensors that could indicate the activation time each time the HSS was used. Thus, the findings cannot provide any insights to understand HH behaviors based on time and use patterns.

For future research, longer duration of HHC monitoring preferably with time-stamped data is recommended. Future research could also examine the influence of unit spatial configurations on other infectious disease control compliance methods in inpatient units. Within the scope of HHC and the effect of spatial configuration, it would be beneficial to develop specific practical guidelines protocols and best practices to assist hospital management in strategic placement of HH resources, and maintaining and upgrading the infrastructure as the process flow changes in the care environment.

The findings from this study, together with future research, could suggest the following implications for practice: the strategic placement of HSSs could potentially increase the cost-effectiveness of HH interventions by increasing the usability of HH resources, and to ensure the adherence to HHC protocols, the cooperation of facility management, architects, and quality control personnel may be needed to ensure the facilities and resources are conducive to HHC.

CONCLUSIONS

Space syntax simulation methods may be a beneficial tool to predict and optimize adherence to HHC. The strategic allocation of HH resources to improve HH in health care environments requires coordination among facility management, health care architects, and quality control personnel and an analytical and objective measurement tool may benefit the process. Improving safety and efficiency in health care requires a systematic approach, taking into account many factors, including the different clinical and nonclinical stakeholders of the health care environment. With improved coordination among multiple stakeholders and disciplines, there is a great potential to improve patient safety and infection control efforts in health care environments.

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APPENDIX

Table A1
Tests of between-subjects effects (model 2)

Dependent variable: Log use

Source	Type III sum of squares	*df*	Mean square	*F*	*P* value
Corrected model	37.736*	5	7.547	13.877	<.001
Intercept	116.493	1	116.493	214.197	<.001
Unit × visibility	1.851	2	0.925	1.702	.190
Unit	.756	2	0.378	0.695	.503
Visibility	22.503	1	22.503	41.377	<.001
Error	35.351	65	0.544		
Total	1,920.409	71			
Corrected total	73.087	70			

*$R^2 = .516$ (adjusted $R^2 = 0.479$).

Table A2
Parameter estimates (model 2)

Dependent variable: Log use

Parameter	β	Standard error	*t*	*P* value	95% Confidence interval	
					Lower bound	Upper bound
Intercept	3.997	0.255	15.658	<.001	3.488	4.507
Unit = cardiac intensive care unit	–0.981	0.300	–3.268	.002	–1.580	–0.382
Unit = Neonatal intensive care unit	0.128	0.260	0.491	.625	–0.392	0.648
Unit = Palliative	0*	–	–	–	–	–
Visibility	0.004	0.001	6.213	<.001	0.003	0.005

*This parameter is set to zero because it is redundant.

Table A3
Tests of between-subjects effects (model 3)

Dependent variable: Log use

Source	Type III sum of squares	*df*	Mean square	*F*	*P* value
Corrected model	30.566*	3	10.189	16.054	<.001
Intercept	17.626	1	17.626	27.773	<.001
Unit	6.957	2	3.478	5.481	.006
Global traffic flow	16.114	1	16.114	25.390	<.001
Error	42.521	67	0.635		
Total	1,920.409	71			
Corrected total	73.087	70			
Corrected model	30.566*	3	10.189	16.054	<.001

*$R^2 = 0.418$ (adjusted $R^2 = 0.392$).

Table A4
Parameter estimates (model 3)

Dependent variable: Log use

Parameter	β	Standard error	*t*	*P* value	95% Confidence interval	
					Lower bound	Upper bound
Intercept	3.103	0.458	6.772	<.001	2.189	4.018
Unit = CICU	–1.184	0.376	–3.150	.002	–1.935	–0.434
Unit = Neonatal intensive care unit	–0.230	0.255	–0.901	.371	–0.740	0.280
Unit = Palliative	0*	–	–	–	–	–
Global traffic flow	0.379	0.075	5.039	<.001	0.229	0.529

*This parameter is set to zero because it is redundant.